

INTERNSHIP PROJECT REPORT

Data Science / Machine Learning Internship

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Organization: upSkill Campus / Uniconverge Technologies (P) Ltd.

Selected Projects:

Project 10 — Quality Prediction in a Mining Process

Project 6 — Remaining Useful Life Prediction for Turbofan Engines

GitHub Repository

<https://github.com/Aryn1102/upskillcampus>

Final consolidated report

1. Internship Overview

The internship provided practical exposure to machine-learning workflows using real-world datasets. The work was organized around two industrial ML problems: quality prediction in a mining flotation process and Remaining Useful Life (RUL) prediction for turbofan engines. The projects were selected to provide experience across data preparation, exploratory data analysis, feature engineering, regression modelling, time-dependent validation, evaluation, and technical documentation.

2. Internship Objectives

- Understand the structure and requirements of real-world machine-learning projects.
- Develop reproducible Python workflows for data loading, preprocessing, analysis, modelling, and evaluation.
- Apply exploratory data analysis and feature engineering to industrial datasets.
- Build baseline regression models and compare their performance using appropriate metrics.
- Understand time-dependent data and predictive-maintenance problems.
- Document implementation, challenges, learnings, and project progress in a structured report.

3. Technology Stack

- **Programming:** Python
- **Data processing:** pandas, NumPy
- **Visualization:** matplotlib

- **Machine learning:** scikit-learn
- **Version control:** Git and GitHub
- **Documentation:** Markdown and PDF reporting

4. Project 10 — Quality Prediction in a Mining Process

4.1 Problem Statement

The project focuses on predicting silica concentration in an iron-ore flotation process. The project specification identifies prediction of concentrate quality as an industrially relevant machine-learning problem because earlier prediction can support corrective process action.

4.2 Data and Initial Analysis

- Loaded the mining-process dataset from CSV format.
- Inspected dataset structure, numerical variables, missing values, and target availability.
- Identified and processed the time variable where available.
- Examined temporal behaviour and relationships between process variables and silica concentration.
- Generated missing-value and correlation outputs for further analysis.

4.3 Preprocessing and Feature Engineering

- Converted suitable object columns to numeric representations.
- Sorted observations chronologically when a timestamp was available.
- Created time-derived features such as hour, day of week, and month.
- Created lagged target features to represent previous process observations.
- Used a chronological train/test split rather than randomly mixing time-dependent observations.

4.4 Machine-Learning Models

- Linear Regression was used as an interpretable baseline.
- Random Forest Regression was used to model nonlinear relationships.
- Gradient Boosting Regression was used as a second nonlinear baseline.
- Median imputation and feature scaling were incorporated through scikit-learn pipelines where appropriate.

4.5 Evaluation

Models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . The implementation saves model metrics and test predictions to an output directory. Numerical scores are intentionally not stated in this report because the final internship dataset and execution environment must be used to generate verified results before submission.

5. Project 6 — Remaining Useful Life Prediction for Turbofan Engines

5.1 Problem Statement

The project addresses predictive maintenance by estimating the remaining operational cycles of turbofan engines from multivariate sensor time-series data. The problem requires modelling degradation over engine cycles rather than treating observations as independent random samples.

5.2 Data Preparation

- Loaded a C-MAPSS-style training file containing engine identifier, cycle, operating settings, and sensor variables.

- Calculated RUL for each observation from the final observed cycle of its engine.
- Created engine-wise rolling sensor statistics to capture local degradation behaviour.
- Kept engine trajectories separated during the train/test split to reduce cross-engine leakage.

5.3 Machine-Learning Models

- Ridge Regression was used as a regularized linear baseline.
- Random Forest Regression was used to capture nonlinear sensor relationships.
- Gradient Boosting Regression was used as another nonlinear baseline.
- Models were evaluated using MAE, RMSE, and R².

5.4 Evaluation and Limitations

The implementation saves model metrics and test predictions for reproducible comparison. The baseline workflow is intended for internship-level experimentation; a production-grade predictive maintenance system would require more rigorous dataset-specific validation, operating-condition analysis, failure-cost-aware metrics, and potentially sequence/deep-learning approaches.

6. Weekly Progress

Week	Main Activities
Week 1	Project understanding, dataset inspection, environment/GitHub setup, initial data-quality analysis, and mode
Week 2	EDA, preprocessing, temporal handling, engine-wise analysis, and initial feature engineering.
Week 3	Baseline regression models, time-aware validation, evaluation metrics, and initial error analysis.
Week 4	Model refinement, explainability-oriented analysis, final evaluation preparation, documentation, and Git organization

7. Challenges Encountered

- Time-dependent datasets require validation strategies that avoid leakage from future observations.
- Industrial sensor and process variables can have different scales, noise levels, and correlations.
- Feature engineering must preserve the temporal structure of the data.
- Predictive-maintenance data requires engine-wise or sequence-wise processing rather than simple row-wisemodelling.
- Model performance must be verified by actually running the final code against the intended internship datasets.

8. Skills and Learning Outcomes

- Improved practical understanding of Python-based ML workflows.
- Applied pandas and NumPy for structured and time-series data processing.
- Applied EDA to understand industrial variables before modelling.
- Gained experience with feature engineering and regression pipelines.
- Learned why chronological and entity-aware validation matters in industrial ML.
- Practiced Git/GitHub-based project organization and reproducible documentation.
- Developed familiarity with industrial quality prediction and predictive-maintenance use cases.

9. Repository and Project Links

GitHub repository: <https://github.com/Aryn1102/upskillcampus>

Project code directory: https://github.com/Aryn1102/upskillcampus/tree/main/project_code

Mining quality prediction code:

https://github.com/Aryn1102/upskillcampus/blob/main/project_code/mining_quality_prediction.py

Turbofan RUL prediction code:

https://github.com/Aryn1102/upskillcampus/blob/main/project_code/turbofan_rul_prediction.py

Weekly reports: <https://github.com/Aryn1102/upskillcampus/tree/main/reports>

10. Conclusion

The internship work provided practical exposure to two complementary industrial machine-learning problems. The mining project developed skills in process-quality prediction and time-aware regression, while the turbofan project introduced predictive maintenance and Remaining Useful Life modelling. Together, the projects provided a foundation in data preparation, exploratory analysis, feature engineering, model development, evaluation, version control, and technical reporting.

Note: This report describes the implemented baseline workflows and documented internship activities. Final quantitative model results should be inserted only after the code has been executed on the final datasets used for submission.